GenreMaster: Genre-Conditioned Audio Enhancement with Differentiable DSP

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***Abstract*—Audio mastering is inherently style dependent: the same spectral, dynamic, and loudness decisions that improve one genre may be inappropriate for another. Existing neural audio enhancement systems rarely model this dependency explicitly, and the lack of large paired mastering datasets makes direct supervised automatic mastering difficult to study. This paper presents GenreMaster, a genre-conditioned audio enhancement framework that combines Transformer-based spectral encoding, feature-wise linear modulation, learnable genre embeddings, and differentiable digital signal processing. Rather than generating audio directly, the model predicts interpretable processing parameters for a mastering-inspired effect chain. We evaluate the framework on GTZAN using controlled synthetic degradation as a proof-of-concept setting. The results show that genre-conditioned transformations can be learned end-to-end and that differentiable DSP provides a useful bridge between neural optimization and audio-engineering structure. The study does not claim production-ready mastering; instead, it identifies a practical path toward genre-aware automatic mastering once suitable paired datasets and perceptual evaluations become available.**

***Index Terms*—audio enhancement, automatic mastering, genre conditioning, differentiable digital signal processing, FiLM conditioning, music information retrieval**

I. Introduction

Audio mastering is the final stage of music production, where a mix is prepared for release through adjustments to loudness, spectral balance, dynamics, and stereo image. These decisions are rarely neutral. A hip-hop master may emphasize low-frequency weight and controlled impact, a classical master may preserve transient detail and dynamic range, and a rock master may favor density without sacrificing intelligibility. As a result, mastering is not only a signal enhancement problem; it is also a style-conditioned decision process.

Recent progress in neural audio processing has produced strong results in enhancement, separation, restoration, and production-oriented modeling. However, most systems treat enhancement as a largely genre-agnostic transformation. This assumption is limiting for music production, where the desired target is shaped by genre conventions, listener expectations, and production aesthetics. A system that optimizes average reconstruction quality may still fail to make style-appropriate processing choices.

Developing automatic mastering systems also faces a data bottleneck. Supervised mastering would ideally require paired examples of unmastered and professionally mastered audio, together with genre or style annotations. Such data is difficult to obtain at scale because professional masters are proprietary and unmastered sources are seldom released alongside final versions. Consequently, research in this area must distinguish between technical feasibility and production readiness. A controlled experiment can test whether a model can learn genre-dependent transformations, but it cannot by itself establish that the model performs professional mastering.

This paper investigates genre-conditioned neural audio enhancement as a step toward automatic mastering. We introduce GenreMaster, a framework that encodes audio features with a Transformer, represents genre labels through learned embeddings, injects those embeddings using feature-wise linear modulation (FiLM), and renders the output through a differentiable digital signal processing (DSP) chain. The model therefore learns to predict parameters for interpretable audio operations rather than directly synthesizing a waveform.

Because large paired mastering datasets are not available, we evaluate GenreMaster in a proof-of-concept setting using GTZAN. We construct synthetic input-target pairs by applying pre-mastering-like degradation to finished audio and training the model to recover the original signal characteristics. This setup is intentionally limited: it tests whether the proposed conditioning and differentiable processing architecture can learn controlled genre-dependent transformations, not whether the system can replace a mastering engineer.

The central question addressed in this paper is: can genre information improve a neural audio enhancement pipeline when the processing itself is constrained to an interpretable differentiable DSP chain? The results indicate that the proposed architecture learns stable transformations across the GTZAN genres, generalizes from validation to test data, and provides a useful experimental foundation for future work with real mastering pairs.

The main contributions of this paper are as follows:

- We present a genre-conditioned audio enhancement framework that combines Transformer spectral encoding, learned genre embeddings, FiLM conditioning, and differentiable DSP.
- We formulate mastering-inspired enhancement as interpretable parameter prediction rather than direct waveform generation.
- We evaluate the framework in a controlled GTZAN proof-of-concept setting with synthetic degradation and stratified genre splits.
- We analyze the gap between controlled reconstruction performance and production-ready automatic mastering, identifying dataset, evaluation, and style-representation requirements for future work.

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This manuscript reports implementation and experiment results from the GenreMaster project.

II. Related Work

A. Automatic Mastering and Neural Audio Enhancement

Automatic mastering aims to approximate the final production decisions normally made by an audio engineer. Commercial services such as LANDR and eMastered indicate practical demand, but academic progress remains constrained by the scarcity of paired unmastered and mastered material. Related research has more often focused on source separation, speech or music enhancement, black-box effect modeling, and automatic mixing [1]–[4].

These tasks share signal-processing components with mastering, but they do not fully address the style-conditioned nature of the mastering target. In particular, a mastering system must decide not only how to make audio cleaner or closer to a reference, but also what kind of spectral and dynamic character is appropriate for the musical context.

B. Genre-Aware Music Modeling

Genre classification has long served as a central task in music information retrieval. GTZAN remains a commonly used benchmark for genre analysis despite known limitations such as fixed-length clips, coarse labels, and dataset artifacts [5]. Modern convolutional and residual encoders can learn discriminative genre representations from spectrograms, making genre labels useful control variables for downstream music-processing systems [6], [7].

Genre conditioning is therefore a natural mechanism for production-aware enhancement. However, genre labels are coarse proxies for style. They capture broad conventions but cannot fully describe artist identity, historical era, arrangement density, or the intended sonic aesthetic. This motivates treating the present work as an initial controlled study rather than a complete model of musical style.

C. Conditioning and Differentiable DSP

Feature-wise linear modulation provides a simple mechanism for injecting conditioning information into neural networks [8]. By learning scale and bias transformations from a control variable, FiLM can adapt intermediate representations without replacing the main encoder.

Differentiable DSP offers a complementary advantage. Instead of asking a network to generate the final waveform directly, differentiable audio effects allow gradient-based training over physically meaningful parameters [9], [10]. For production tasks, this interpretability is important: equalization, compression, limiting, and stereo width are familiar operations with constraints that are easier to inspect than unconstrained waveform synthesis. Multi-resolution spectral losses further support perceptually relevant optimization in audio generation and enhancement [11], [12].

GenreMaster integrates these ideas into a mastering-inspired pipeline. The novelty is not any single component in isolation, but the way genre conditioning, Transformer encoding, and differentiable DSP are combined to study style-dependent audio enhancement under a controlled experimental protocol.

III. Methodology

A. Problem Formulation

Let x denote an input audio signal and let g denote its genre label. The objective is to learn a transformation f_θ that produces an enhanced signal $\hat{y}$:

$$\hat{y} = f_\theta(x, g). \tag{1}$$

In GenreMaster, this transformation is decomposed into audio encoding, genre conditioning, DSP-parameter prediction, and differentiable rendering. Rather than modeling $\hat{y}$ as an unconstrained neural waveform, the system predicts a parameter vector ϕ for a mastering-inspired processing chain:

$$\phi = P_\theta(E_\theta(x), z_g), \tag{2}$$

$$\hat{y} = D_\phi(x), \tag{3}$$

where E_θ is the audio encoder, z_g is the learned genre embedding, P_θ is the FiLM-conditioned parameter predictor, and D_ϕ is the differentiable DSP chain.

This formulation is designed to preserve a connection between the learned model and audio-engineering practice. The network learns how strongly to apply processing, while the rendering stage remains structured around recognizable audio operations.

B. Architecture

The architecture consists of four components, shown in Fig. 1. First, log-mel spectral features are extracted from the input audio and passed to a Transformer encoder [13]. The encoder models temporal dependencies in the spectral representation and produces a compact audio feature vector.

Second, each genre label is mapped to a learned embedding. This embedding represents genre as a continuous conditioning signal rather than as a hard-coded rule set. Third, the genre embedding modulates the audio representation using FiLM:

$$\mathrm{FiLM}(h, z_g) = \gamma(z_g) \odot h + \beta(z_g), \tag{4}$$

where h is the encoded audio representation and $\gamma(\cdot)$ and $\beta(\cdot)$ are learned affine transformations of the genre embedding.

Finally, the conditioned representation is mapped to parameters for a differentiable DSP chain. The chain includes loudness control, parametric equalization, multiband compression, limiting, and stereo-width adjustment. These modules are selected because they correspond to common mastering operations while remaining amenable to differentiable optimization.

C. Training Objective

Training minimizes a weighted combination of loudness, spectral, dynamic, and perceptual terms:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{loud}}\mathcal{L}_{\text{loud}} + \lambda_{\text{spec}}\mathcal{L}_{\text{spec}} + \lambda_{\text{dyn}}\mathcal{L}_{\text{dyn}} + \lambda_{\text{perc}}\mathcal{L}_{\text{perc}}. \tag{5}$$

The loss design reflects the fact that mastering quality cannot be captured by a single waveform distance. Loudness alignment, spectral balance, dynamic behavior, and perceptual similarity each capture a different aspect of the desired transformation. In the present proof-of-concept, the loss is used to test learnability under controlled conditions rather than to define a complete perceptual mastering criterion.

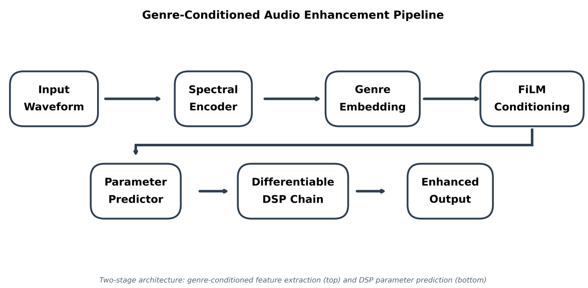

Fig. 1. GenreMaster pipeline. Audio features are encoded, modulated by learned genre embeddings, and mapped to parameters for a differentiable mastering-inspired DSP chain.

D. Data and Experimental Setup

The experiment uses GTZAN after removing one corrupted file, yielding 999 valid clips across ten genres. The data are split in a stratified manner into training, validation, and test partitions, as summarized in Table I. All audio is resampled to 22,050 Hz and represented as 30-second clips.

Because GTZAN contains finished music rather than unmastered/mastered pairs, we construct synthetic training pairs. The input is produced by applying degradation operations that imitate some pre-mastering deficiencies, including reduced loudness, spectral imbalance, and altered dynamics. The original clip is then used as a pseudo-target. This procedure provides a controlled learning problem but should not be interpreted as a faithful simulation of professional mastering.

TABLE I
DATASET SPLIT CONFIGURATION

Set	Clips	Notes
Training	799	Stratified by genre
Validation	99	Stratified by genre
Test	101	Stratified by genre

TABLE II
TRAINING CONFIGURATION

Component	Value
Sample rate	22,050 Hz
Clip duration	30.0 s
Batch size	8
Gradient accumulation	4
Optimizer	AdamW [14]
Learning rate	3×10^{-4}
Weight decay	1×10^{-4}
Scheduler	Cosine with warmup
Gradient clipping	1.0
Device backend	CUDA

IV. RESULTS

A. Training Behavior

Fig. 2 shows the training and validation trajectories. The model converges rapidly and maintains a small gap between training and validation loss. This behavior suggests that the architecture is able to learn the controlled degradation-reversal task without severe overfitting.

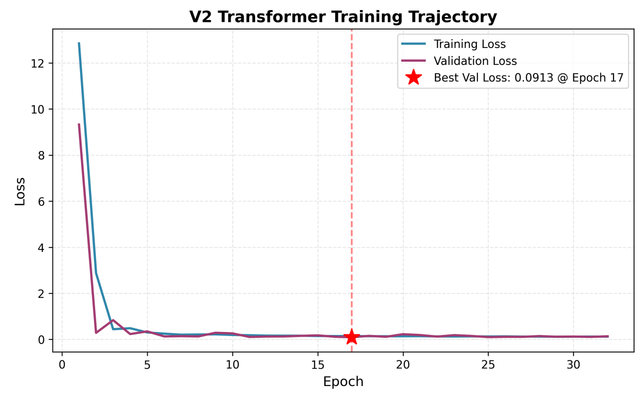

Fig. 2. Training and validation loss curves for GenreMaster on the controlled GTZAN enhancement task.

B. Test Performance

Table III reports the held-out test results. The test loss closely matches the best validation loss, and waveform-level metrics indicate strong reconstruction quality in the synthetic setting. These values should be interpreted as evidence that the architecture learns the controlled transformation, not as evidence of professional mastering quality.

TABLE III
HELD-OUT TEST PERFORMANCE

Metric	Value
Test samples	101
Validation loss	0.0913
Test loss	0.0903
SNR	16.67 dB
Correlation	0.9892
Best epoch	17

C. Genre-Dependent Behavior

Fig. 3 reports test loss by genre. The losses remain within a narrow range across the ten GTZAN categories. This consistency suggests that genre conditioning does not simply benefit one or two genres while failing on others. Instead, the framework learns a shared enhancement structure that can be adapted through the genre embedding.

D. Auxiliary Genre Separability

To verify that the dataset supports meaningful genre-conditioned modeling, we trained a ResNet-based genre classifier on the same corpus. The classifier reached high validation accuracy, as shown in Fig. 4, indicating that the genre labels are learnable from the spectral representation. This result supports the use of genre as a conditioning variable, while also reinforcing that GTZAN genre labels are coarse and should not be treated as complete style descriptors.

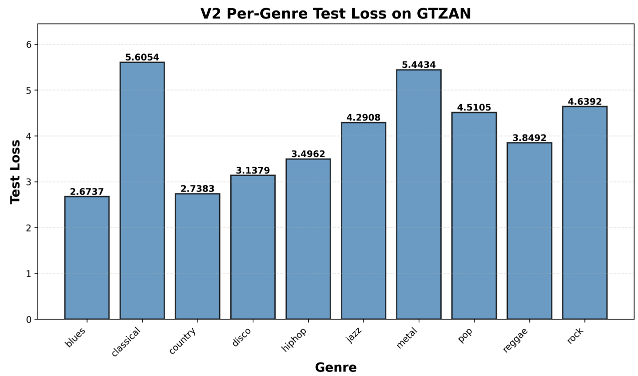

Fig. 3. Per-genre test loss on GTZAN. The narrow spread indicates stable behavior across the evaluated genre labels.

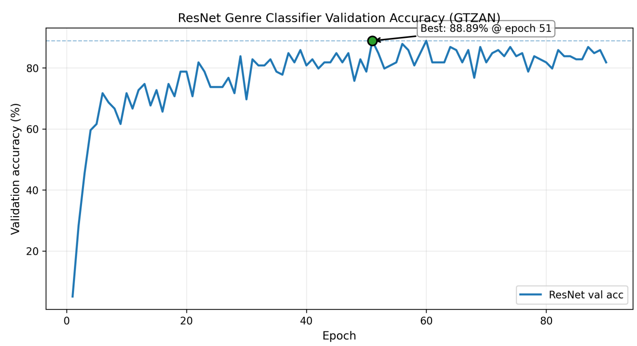

Fig. 4. Validation accuracy trajectory for the auxiliary ResNet genre classifier on GTZAN.

V. Discussion

A. What the Experiment Shows

The results show that a neural system can learn genre-conditioned audio transformations when the output space is constrained through differentiable DSP. The small validation-test gap indicates stable generalization within the controlled GTZAN setting. The per-genre analysis further suggests that the model does not rely on a single global correction but adapts its processing across genre labels.

The architecture also provides a practical advantage over direct waveform generation. Because the model predicts parameters for familiar audio effects, its behavior can be inspected in terms closer to mastering practice. This does not make the system equivalent to a mastering engineer, but it gives the learning problem a useful structure and reduces the opacity of the output transformation.

B. What the Experiment Does Not Show

The main limitation is the absence of real mastering pairs. GTZAN contains already-finished music, and the degraded inputs are generated synthetically. Real pre-mastering material differs in recording quality, mix balance, headroom, noise, room coloration, arrangement density, and artistic intent. A model trained only to reverse synthetic degradation may learn useful signal transformations without learning the judgment required for professional mastering.

The evaluation metrics are also limited. Reconstruction loss, correlation, and SNR are useful for controlled experiments, but mastering is ultimately perceptual and contextual. A technically close reconstruction is not necessarily a better master. Future evaluations should include loudness and true-peak compliance [15], perceptual audio metrics [16], [17], comparisons with commercial systems, and listening studies involving trained engineers.

Finally, genre labels are an imperfect representation of style. Production conventions vary within genres and evolve over time. A more complete system should learn continuous style embeddings that can capture subgenre, era, artist preference, and reference-track intent.

C. Path Toward Automatic Mastering

The present study suggests three requirements for moving from proof-of-concept enhancement to automatic mastering. First, the field needs paired datasets containing unmastered and professionally mastered versions across diverse genres and production contexts. Second, models should be evaluated with perceptual and preference-based criteria rather than reconstruction metrics alone. Third, conditioning should move beyond discrete genre labels toward richer style representations that can reflect real mastering goals.

Under these conditions, the GenreMaster architecture could serve as a foundation for a more realistic system: the Transformer encoder would model musical context, the conditioning pathway would represent target style, and the differentiable DSP chain would provide controllable, interpretable rendering.

VI. Conclusion

This paper presented GenreMaster, a genre-conditioned audio enhancement framework that combines Transformer spectral encoding, FiLM conditioning, learned genre embeddings, and differentiable DSP. The model predicts interpretable parameters for a mastering-inspired processing chain rather than generating audio directly.

In a controlled GTZAN proof-of-concept experiment, GenreMaster learned stable degradation-reversal transformations and generalized consistently across held-out clips and genre categories. These findings support the feasibility of genre-conditioned differentiable DSP for music enhancement.

At the same time, the study is intentionally bounded. The experiment does not establish production-ready automatic mastering because it relies on synthetic degradation and finished music rather than true unmastered/mastered pairs. Future work should focus on real mastering datasets, perceptual evaluation, richer style conditioning, and comparisons with professional and commercial mastering systems.

Ethics and Reproducibility Statement

All experiments used publicly available music data and code-executed metrics from repository artifacts. No human subject data were collected. The reported values are derived from saved experimental outputs, and the limitations of the proof-of-concept evaluation are stated explicitly.

References

[1] A. Defossez, G. Synnaeve, and Y. Adi, "Real time speech enhancement in the waveform domain," in *Proceedings of Interspeech*, 2020, pp. 3291–3295.

[2] A. Mäkitalo, S. Disch, and V. Pulkki, "Deep learning for black-box modeling of audio effects," in *Applied Sciences*, vol. 10, no. 2, 2020, p. 638.

[3] C. J. Steinmetz, J. Pons, S. Pascual, and J. Serra, "Automatic multitrack mixing with a differentiable mixing console of neural audio effects," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021, pp. 71–75.

[4] G. P. Colón and G. De Poli, "Artificial intelligence for music production," *IEEE Signal Processing Magazine*, vol. 38, no. 4, pp. 41–55, 2021.

[5] G. Tzanetakis and P. Cook, "Musical genre classification of audio signals," *IEEE Transactions on Speech and Audio Processing*, vol. 10, no. 5, pp. 293–302, 2002.

[6] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[7] S. Gururani and A. Lerch, "Instrument identification in polyphonic music using convolutional neural networks," in *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*, 2019, pp. 307–314.

[8] E. Perez, F. Strub, H. De Vries, V. Dumoulin, and A. Courville, "Film: Visual reasoning with a general conditioning layer," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018, pp. 3942–3951.

[9] J. Engel, L. Hantrakul, C. Gu, and A. Roberts, "Ddsp: Differentiable digital signal processing," in *International Conference on Learning Representations (ICLR)*, 2020.

[10] M. A. M. Ramírez and J. D. Reiss, "Differentiable signal processing with black-box audio effects," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021, pp. 171–175.

[11] C. J. Steinmetz and J. D. Reiss, "auraloss: Audio focused loss functions in pytorch," in *Digital Music Research Network One-day Workshop (DMRN+15)*, 2020.

[12] R. Yamamoto, E. Song, and J.-M. Kim, "Parallel wavegan: A fast waveform generation model based on generative adversarial networks with multi-resolution spectrogram," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020, pp. 6199–6203.

[13] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[14] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.

[15] ITU-R, "Algorithms to measure audio programme loudness and true-peak audio level," International Telecommunication Union, Tech. Rep. BS.1770-4, 2015.

[16] K. Kilgour, M. Zuluaga, D. Roblek, and M. Sharifi, "Fréchet audio distance: A reference-free metric for evaluating music enhancement algorithms," in *Proceedings of Interspeech*, 2019.

[17] P. Manocha, A. Finkelstein, R. Zhang, N. J. Bryan, G. J. Mysore, and Z. Jin, "Cdpam: Contrastive learning for perceptual audio similarity," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021, pp. 196–200.